Financial Innovation

RESEARCH Open Access

Machine learning-based portfolio optimization: comparative analysis with the all-weather portfolio strategy

Yu Sung Ha[1], Jongho Kang[2], Jihun Kim[3] and Dohyun Chun[4*]

*Correspondence:
dohyunc@kangwon.ac.kr

[1] School of Accounting and Finance, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong, China
[2] College of Business Administration, Chonnam National University, 77, Yongbong-Ro, Buk-Gu, Gwangju, Republic of Korea
[3] College of Humanities & Social Sciences Convergence, Yonsei University (Mirae Campus), 1, Yonseidae-Gil, Wonju-Si, Gangwon-Do, Republic of Korea
[4] College of Business Administration, Kangwon National University, 1, Gangwondaehak-Gil, Chuncheon-Si, Gangwon-Do, Republic of Korea

Abstract

This study investigates whether machine learning effectively processes high-dimensional data, a challenging task for traditional predictive models, to optimize portfolio strategies. Using daily data from December 2004 to July 2024, we compare various machine-learning models for asset allocation in an all-weather portfolio comprising exchange-traded funds for the S&P 500, long-term Treasury bonds, and gold. We find that the LASSO and elastic net models exhibit superior overall performance, whereas tree-based models excel in forecasting long-term Treasury bond returns. Portfolio strategies employing these models achieve Sharpe ratios near 0.70, substantially outperforming static benchmarks. The results demonstrate that machine learning can optimize portfolio performance in practical investment settings.

Keywords: Market prediction, Portfolio diversification, Machine learning, All-weather portfolio, Asset allocation

JEL Classification: C55, G11, G17

Introduction

Machine learning has garnered significant attention in financial modeling due to its ability to capture high-dimensional interactions and complex nonlinearities that traditional approaches often overlook. However, despite its growing role in financial decision-making over the past decade, translating its technical advantages into practical investment strategies remains elusive. This gap between theoretical expectations and realized outcomes is often attributed to predictive models' inability to fully consider various constraints encountered during actual trading. Among these challenges, our study leverages exchange-traded funds (ETFs), with their high liquidity, low transaction costs, and inherent diversification, to assess the practical viability and limitations of machine learning-based investment strategies.

Although accurate forecasts of asset returns are essential for strategy design, translating these signals into realized outcomes is contingent upon implementation feasibility. Market frictions, such as transaction costs, bid-ask spreads, and price impacts, can erode expected returns, creating a marked discrepancy between projected and attainable

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

investor returns (Avramov et al. 2023). Furthermore, managing a universe of hundreds of individual assets introduces fundamental operational complexity. As the portfolio size increases, integrating diverse liquidity conditions and practical constraints across individual assets into the modeling becomes a formidable task. An investment environment that mitigates the complexities of individual asset management is necessary to realize the benefits of machine learning effectively.

From this perspective, using ETFs to assess machine learning applications in portfolio management has several advantages. By combining entire asset classes into one inexpensive trading vehicle, ETFs allow for direct exposure adjustments through unified transactions rather than individual constituent management (Glosten et al. 2021). For short-horizon investors using dynamic allocation strategies, deep secondary market liquidity in major ETFs provides a competitive edge that allows frequent portfolio adjustment (Ben-David et al. 2018; Khomyn et al. 2024). Moreover, owing to minimal tracking errors and tax-efficient structures that protect unrealized gains, operating costs remain consistently lower than those of alternative vehicles.

We employ Dalio's (2017) all-weather portfolio as a benchmark to examine the validity of machine learning-based dynamic asset allocation strategies using ETFs' practical advantages. The all-weather portfolio is widely recognized for its stable performance across diverse market regimes. Its underlying philosophy aligns with machine learning-based portfolio strategies, as both aim to effectively adapt to changing market conditions. Moreover, the all-weather portfolio operates as an ETF-based strategy in practice, which allows for direct and realistic comparison. Following the all-weather portfolio framework, we select three core assets that respond to different macroeconomic conditions—equities for growth, long-term bonds for deflation, and gold for crisis hedging.

We apply machine learning models to build portfolios based on daily return forecasts for core asset classes. In particular, we predict the daily returns of ETFs tracking these assets—SPY (S&P 500 index), TLT (U.S. long-term Treasury bonds), and GLD (gold)—using 47 predictive variables, including U.S. economic indicators, stock market metrics, and global indices from December 2004 to July 2024. As the data are high-dimensional, we use a set of machine learning models: penalized regression methods, including ridge, LASSO, and elastic net (ENet); tree-based models, including random forest (RF), light gradient boosting machine (LGBM), and extreme gradient boosting (XGB); and deep learning models.

Our findings indicate that LASSO and ENet consistently outperform other models in predicting the S&P 500, long-term Treasury bonds, and gold. Based on the out-of-sample coefficient of determination, LASSO and ENet deliver the highest predictive accuracy for S&P 500 index returns, with R^2_{oos} values of 3.75% and 3.60%, respectively. For gold, LGBM achieves the highest accuracy, with linear model-based machine learning showing comparable performance. Tree-based models excel for long-term Treasury bonds, while artificial neural networks show weak predictive power across all assets.

These predictive advantages extend to portfolio performance. The LASSO-based strategy, with the highest predictive accuracy, yields an average annual return of 13.24% and a Sharpe ratio of 0.71. ENet follows closely, with a 12.90% annual return and a Sharpe ratio of 0.69. In contrast, all three static asset allocation strategies using the all-weather portfolio yield up to 7.00% annually, with Sharpe ratios limited to 0.58. Among simple index

investment strategies, the SPY performs best, averaging 10.56% annually with a Sharpe ratio of 0.47. These results remain robust even after accounting for transaction costs, as LASSO and ENet maintain higher Sharpe ratios than the established benchmarks.

This study contributes to the emerging literature on machine learning-based portfolio strategies in two ways. First, we reassess the effectiveness of these strategies by using ETFs as practical, cost-efficient investment vehicles. While numerous studies have examined asset allocation strategies across large sets of individual securities (e.g., Gu et al. 2020), our approach achieves effective diversification with a limited number of ETFs under less restrictive and more realistic conditions. Second, we compare static and dynamic allocation strategies against the widely adopted all-weather portfolio. This realistic benchmark highlights the practical role of machine learning in improving asset allocation beyond simple statistical differences between models. Overall, our findings substantiate the advantages of machine learning-based strategies over traditional approaches in today's complex, fast-changing market environment.

The remainder of this paper is structured as follows. Sect. "Related Literature" reviews the related literature on machine learning in finance. Sect. "Methodology" introduces the machine learning models and portfolio optimization methods used in this study. Sect. "Empirical analysis" presents the empirical analysis, including prediction accuracy and portfolio performance results. Sect. "Conclusion" presents the conclusion.

Related literature

Early research on portfolio allocation is based on Markowitz's (1952) mean–variance optimization framework, which formalizes the risk–return trade-off and lays the theoretical foundation of modern portfolio theory. However, despite its theoretical appeal, implementing mean–variance optimization poses several challenges. Frankfurter et al. (1971) document that mean–variance optimized portfolios are highly sensitive to estimation errors. Michaud (1989) and Chopra and Ziemba (1993) further argue that even small changes in input parameters can lead to substantial variations in portfolio composition and performance. To address these limitations, various advanced techniques have been proposed, including shrinkage methods (Jorion 1986), Bayesian frameworks that incorporate investor views (Black and Litterman 1992), and practical portfolio constraints (Jagannathan and Ma 2003). However, despite these advances, DeMiguel et al. (2009) highlight that traditional optimization techniques often fail to outperform simple heuristic strategies such as the naïve equally weighted allocation consistently.

Therefore, researchers have increasingly embraced machine-learning techniques for return prediction and asset allocation. Regularized linear models, such as ridge (Hoerl and Kennard 1970) and LASSO (Tibshirani 1996), shrink or select coefficients to control variance and mitigate multicollinearity, with elastic net (Zou and Hastie 2005) combining L_1 and L_2 penalties to provide a more stable solution when variables are highly correlated. Empirically, Rapach et al. (2013) demonstrate that the adaptive elastic net is useful for estimating high-dimensional predictive models of international equity returns. Kozak et al. (2020) highlight the risk of overfitting in high-dimensional settings, showing the efficacy of regularized linear models in constructing the stochastic discount factor. Similarly, Freyberger et al. (2020) reveal that a LASSO-based framework shows good performance for high-dimensional characteristic selection.

Tree-based techniques address different margins by capturing nonlinearities and interactions. Gu et al. (2020) demonstrate that random forests and gradient-boosted regression trees outperform linear models in cross-sectional return prediction and enhance portfolio outcomes, partly because they flexibly model nonlinear interactions among predictors. Bryzgalova et al. (2023) introduce tree-based portfolio construction, effectively capturing complex, nonlinear interactions among firm characteristics. Broadly, tree-based models demonstrate robust performance in delinquency prediction, systemic risk applications (Khandani et al. 2010), and credit default prediction (Alonso Robisco and Carbó 2022).

As data become richer and computationally cheaper, more complex models have gained traction in empirical practice. Deep learning, in particular, employs multilayer neural networks to compose nonlinear transformations into hierarchical representations, enabling more flexible, data-driven modeling of high-dimensional, heterogeneous signals. Feng et al. (2018) document the effectiveness of neural networks in predicting the equity market risk premium. Gu et al. (2020) and Wang (2024) use neural network models to study the cross-section of stock returns, highlighting the potential of deep learning in empirical asset pricing. Beyond stock markets, Foroutan and Lahmiri (2024) demonstrate that these models perform well in forecasting major commodity prices, including crude oil, gold, and silver. Moreover, whereas supervised learning depends on labeled data, Cong et al. (2021) reveal the benefit of deep reinforcement learning when it directly optimizes portfolio objectives by interacting with the environment. Jiang and Liang (2017) apply deep reinforcement learning to cryptocurrency markets, directly outputting portfolio weights without price prediction, and show strong empirical results in managing portfolios.

Building on these advances, Vaswani et al. (2017) introduce the self-attention mechanism, opening new horizons in natural language processing and leading studies to adopt transformer-based approaches. For example, Lim et al. (2021) show strong performance in forecasting the future volatility of stock indices by combining attention mechanisms with interpretable components in a multi-horizon setting. Wang et al. (2022) and Zhang et al. (2022) apply transformer models to stock market prediction and report significant improvements over traditional methods. In addition, machine learning finds applications in volatility timing and regime-dependent portfolio choice (Chen et al. 2024; Cong et al. 2021; Lim et al. 2021).

The increasing role of machine learning in finance stems from several factors. First, machine learning effectively handles high-dimensional data challenges through nonlinear dimensionality reduction and principal component analysis, extracting meaningful insights from complex data structures (Freyberger et al. 2020; Gu et al. 2021; Kelly et al. 2023, 2019; Lettau and Pelger 2020; Yan and Yu 2023). Second, machine learning overcomes the limitations of traditional econometric models that assume linear relationships between dependent and explanatory variables, enabling accurate return prediction and improved portfolio performance (Chun et al. 2024; Feng et al. 2024; Gu et al. 2020; Htun et al. 2023). Consistently, a growing body of empirical evidence demonstrates the effectiveness of machine learning within various financial and economic contexts (Ban et al. 2018; Chinco et al. 2019; Martin and Nagel 2022; Rapach et al. 2013; Awan et al. 2021).

However, translating such technical advantages into actionable investment strategies remains a challenge, particularly with real-world trading frictions. Recent research has addressed these concerns by directly incorporating turnover penalties, transaction costs, and rebalancing constraints into training and optimization processes, finding that meaningful alpha can survive even after accounting for such costs (e.g., Aleti et al. 2025; Freyberger et al. 2020; Leippold et al. 2022). At the same time, emerging evidence highlights the fragility of machine learning–based strategies in live trading environments. For example, Avramov et al. (2023) find that machine learning portfolios may suffer performance losses in the presence of trading costs due to high turnover or extreme positions. Similarly, Wolff and Echterling (2024) note that the practical viability of machine learning-based strategies requires efficient trading with low transaction costs.

Methodology

Machine learning models

Linear models

Traditional financial models use ordinary least squares (OLS) regression, which assumes a linear relationship between the dependent and explanatory variables. Given an estimated coefficient vector β, OLS minimizes the sum of squared residuals (L_2) as follows:

$$b = argmin_\beta \frac{1}{T}\sum_{t=1}^{T}\left(y_{t+1} - X_t\beta\right)^2, \tag{1}$$

where T represents the number of observations, y is the dependent variable, $X_t = (X_{1,t}, X_{2,t}, \ldots, X_{R,t})$ and $\beta = (\beta_1, \ldots, \beta_R)'$ denote the independent variable and parameter vector, respectively, and R is the number of independent variables. However, simple linear models are ineffective for high-dimensional data analysis. As dimensionality increases, data density decreases, weakening the OLS approach, which minimizes the distance between the regression line and observed data points. To address overfitting, the number of estimated parameters must be reduced. Unlike traditional models, machine learning mitigates overreliance on specific independent variables by applying regularization, which stabilizes out-of-sample predictions while accommodating large datasets. Ridge regression introduces a penalty term ν to minimize estimation errors in OLS, resulting in the following objective function:

$$b^r = argmin_\beta \frac{1}{T}\sum_{t=1}^{T}\{\left(y_{t+1} - X_t\beta\right)^2 + \nu\|\beta\|_2^2\} \tag{2}$$

Where $\|\beta\|_2^2$ represents the sum of squares of the parameters. Ridge regression balances bias and variance by adding shrinkage regularization to the L_2 norm. This reduces overfitting by driving insignificant variable coefficients closer to zero.

LASSO regression applies a different penalty function than ridge regression. Its objective function incorporates an L_1-norm penalty:

$$b^l = argmin_\beta \frac{1}{T}\sum_{t=1}^{T}\{\left(y_{t+1} - X_t\beta\right)^2 + \nu\|\beta\|_1\} \tag{3}$$

where $\|\beta\|_1$ represents the sum of absolute parameter values. Unlike ridge regression, LASSO enables feature selection by setting coefficients of insignificant variables to zero, further enhancing model interpretability.

Elastic Net regression integrates L_1 and L_2-norm penalties to address limitations in standard regression models. The objective function is defined as follows:

$$b^e = argmin_\beta \frac{1}{T}\sum_{t=1}^{T}\{\left(y_{t+1} - X_t\beta\right)^2 + \nu(1-\lambda)\|\beta\|_1 + \nu\lambda\|\beta\|_2^2\} \tag{4}$$

When $\lambda = 0$, elastic net regression is equivalent to LASSO regression. When $\lambda = 1$, it corresponds to ridge regression.

Tree-based models

As previously discussed, penalized linear regression mitigates overfitting in ordinary least squares (OLS) by constraining coefficients. However, it remains limited in capturing nonlinear relationships and interactions among predictive variables. Tree-based models address these limitations in three ways. First, unlike linear models, which require normal residual distributions, tree-based models impose no assumptions on the data distribution. Second, tree-based models identify complex interactions by iteratively splitting features based on predefined criteria. Third, tree-based models apply to both regression (predicting continuous values) and classification (categorizing data) tasks.

The decision tree is the foundational tree-based model. It employs a tree-like structure that recursively partitions data into child nodes based on optimal splitting criteria. The objective is to maximize homogeneity within each split, measured using impurity metrics such as the Gini index, entropy, and chi-square statistic. Decision trees evolved into random forest and gradient boosting, which were further refined into LGBM and XGB. Random forest ensembles multiple independent decision trees, reducing overfitting and improving predictive accuracy. Gradient boosting, similar to random forest, uses multiple decision trees, with each new tree minimizing the loss function of its predecessor. LGBM modifies gradient boosting by employing leaf-wise splitting, allowing asymmetric tree structures. XGB extends gradient boosting with L_1 and L_2-norm penalties. This study applies random forest, LGBM, and XGB.

Artificial neural networks

Artificial neural networks (ANNs) simulate the structure and function of biological neural networks to train computers. This approach, also known as deep learning, builds on neuron-like computational units. The feed-forward neural network is the simplest artificial neural network model, comprising an input layer for input variables, hidden layers that learn interactions and nonlinear structures, and an output layer that synthesizes signals from these hidden layers to produce final results. This learning method uses given weight values for each layer to derive final results in the output layer and then adjusts weights to reduce the error between actual values and output values through the backpropagation method. This process continues until the model achieves an acceptable error threshold. This study employs the multilayer perceptron (MLP) architecture, with up to five hidden layers.

Hyperparameter optimization

This study forecasts daily returns for three ETFs—SPY (S&P 500), TLT (Treasury bonds), and GLD (gold). Basically, the function predicting future returns can be expressed as $\widehat{r}^{m}_{t+1} = g^{m}(X_t)$, where $g^{m}(\cdot)$ represents the function derived through the models mentioned above. To determine the optimal configuration for each model, we utilize a rolling window approach where parameters are estimated on a training set of 1,000 daily observations and subsequently evaluated on a 250-day validation set. Within this framework, we perform a grid search over the parameter search spaces detailed in Appendix Table A1 to identify the optimal parameters that minimize the mean squared error (MSE) within the validation window. This sequential split validation is strictly designed to ensure that no information from future observations contaminates parameter estimation. The process is updated annually in June, as we found negligible improvements from more frequent retraining. Consequently, the out-of-sample predictions cover a period of 4,165 days, spanning from December 1, 2007, to July 24, 2024. The optimal hyperparameters selected for each estimation period are reported in Appendix Table A2.

Out-of-sample prediction

We use MSE and the out-of-sample coefficient of determination (R^2_{oos}) to evaluate the accuracy of out-of-sample predictions. MSE is defined as the sum of squared differences between actual returns and predicted returns. R^2_{oos} is calculated as follows:

$$R^2_{oos} = 1 - \frac{\frac{1}{T_p}\sum_{t\in \mathrm{P}}\left(r_t - \widehat{r}_t\right)^2}{\frac{1}{T_p}\sum_{t\in \mathrm{P}}\left(r_t - \overline{r}_t\right)^2}, \tag{5}$$

where P represents the prediction point, T_p represents the prediction period, and $\overline{r}_t$ represents the 20-day moving average. To compare model predictive performance, we apply the Diebold and Mariano (1995, DM) test, calculated as follows:

$$K_{m_1,m_2} = \frac{\overline{k}_{m_1,m_2}}{se\left(k_{m_1,m_2}\right)}, \quad k_{m_1,m_2} = \frac{1}{T_p}\sum_{t\in \mathrm{P}}\{\left(\hat{r}^{m_2}_t - r_t\right)^2 - \left(\hat{r}^{m_1}_t - r_t\right)^2\} \tag{6}$$

where $\overline{k}$ and $se(k)$ represent the mean and standard deviation of k, respectively. Moreover, k_{m_1,m_2} shows the relative difference in how much models m_1 and m_2 deviate from actual values during the prediction period. Essentially, if K_{m_1,m_2} has a significant positive value, model m_1 makes more accurate predictions than model m_2. Conversely, if it has a negative value, model m_2 makes more accurate predictions than model m_1. Finally, we use the historical mean, defined as the average daily return from the previous month, as a benchmark. Figure 1 summarizes the methodological framework.

Empirical analysis

Data

This study utilizes daily data from December 1, 2004, to July 24, 2024 (5,126 observations), covering 44 financial and economic indicators plus three ETFs—SPY (SPDR S&P 500 ETF Trust), TLT (iShares 20 + Year Treasury Bond ETF), and GLD (SPDR Gold Shares). While the original all-weather portfolio includes commodities, we exclude them due to their substantial underperformance during the sample period, which consistently

Table 1 Descriptive statistics

	SPY	TLT	GLD
Panel (A) descriptive statistics			
Mean	0.05	0.02	0.04
Median	0.07	0.04	0.05
Max	14.52	7.52	11.29
Min	−10.94	−6.67	−8.78
Std. Dev	1.20	0.93	1.11
Skewness	−0.07	0.09	−0.18
Kurtosis	15.02	3.54	6.21
ADF	0.00	0.00	0.00
Panel (b) correlation coefficients			
SPY	–	−0.33	0.05
TLT	−0.33	–	0.17
GLD	0.05	0.17	–

This table provides descriptive statistics for the key variables in this study: SPY, TLT, and GLD. Panel (a) reports the mean, median, maximum, minimum, standard deviation (Std. Dev.), skewness, kurtosis, and *p*-value of the Augmented Dickey Fuller (ADF) test over the sample period. Panel (b) presents the correlation coefficients between these variables. The sample period extends from December 1, 2004, to July 24, 2024

Table 2 Out-of-sample prediction performance

	Panel (a) SPY		Panel (b) TLT		Panel (c) GLD	
	MSE	R^2_{oos}(%)	MSE	R^2_{oos}(%)	MSE	R^2_{oos}(%)
Ridge	1.34	−5.84	1.04	−7.04	1.14	−3.28
LASSO	1.27	3.75	0.99	2.32	1.10	4.45
ENet	1.28	3.60	0.99	2.32	1.10	4.41
RF	1.30	0.04	0.99	2.13	1.14	−1.95
XGB	1.28	2.68	0.99	3.57	1.11	2.34
LGBM	1.28	2.56	0.99	3.31	1.10	4.51
MLP1	1.53	−38.68	1.22	−48.18	1.48	−73.42
MLP2	1.42	−18.69	1.09	−18.84	1.27	−26.93
MLP3	1.33	−4.82	1.02	−3.11	1.19	−11.85
MLP4	1.34	−5.73	1.02	−2.40	1.16	−5.59
MLP5	1.30	−0.10	1.02	−2.47	1.11	2.85
OLS	1.34	−6.25	1.05	−8.46	1.15	−3.79
HM	1.30	–	1.00	–	1.12	–

This table presents the performance of the prediction models for the S&P 500, long-term Treasury bonds, and gold returns. The mean squared error (MSE) represents the average squared difference between the predicted and actual values. R^2_{oos} is the out-of-sample coefficient of determination

deteriorates portfolio performance across all strategies. Our sample period deliberately includes major market crises—the 2008 financial crisis, the European debt crisis, and the COVID-19 pandemic—ensuring that the results reflect various market conditions. The predictive variables comprise U.S. economic indicators (credit spreads, inflation expectations, interest rates, and economic surprises), equity market measures (VIX, sector indices, and sentiment indicators), foreign exchange rates (Dollar Index and major currency pairs), and global market indices (international equities, commodities, and economic surprises). All variables are observed at daily frequency in their original form, with detailed definitions provided in Appendix Table A3. We apply the Yeo–Johnson power transformation (Yeo and Johnson 2000) to standardize variables, estimating

parameters on the training set only to prevent data leakage. Combined with lagged returns of the three ETFs, this yields 47 input variables. Table 1 presents descriptive statistics and correlations for the three target variables over the sample period.

Out-of-sample prediction performance

This section assesses the predictive performance of the machine learning models. Table 2 presents results using MSE and R^2_{oos} as indicators of prediction accuracy, as mentioned in Sect. "Hyperparameter optimization". Panels (a), (b), and (c) show the prediction accuracy for S&P 500, long-term Treasury bonds, and gold ETF returns, respectively. Moreover, LASSO and ENet provided the highest accuracy for the S&P 500 index. These models exhibited the lowest error (MSE values of 1.27 and 1.28, respectively) and the highest explanatory power, with R^2_{oos} values of 3.75% and 3.60%, respectively. Conversely, OLS performed worse than the historical mean. Tree-based models—random forest, LGBM, and XGB—delivered the best results for long-term Treasury bond returns, with similar performance across these models. For artificial neural networks, performance

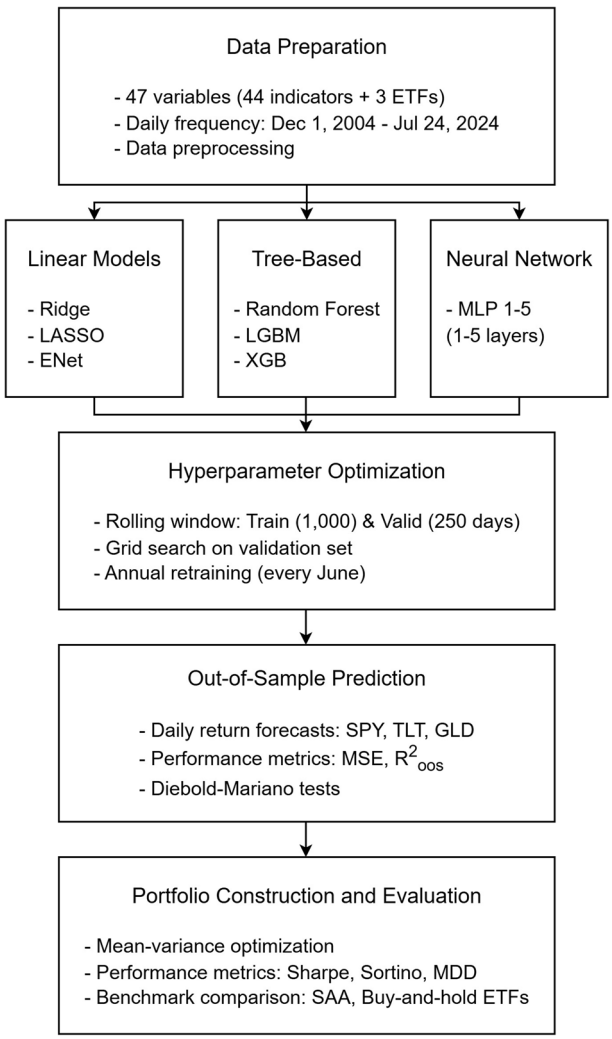


Fig. 1 Research framework

was poor across all assets. In particular, for the S&P 500, the models produced negative R^2_{oos}, indicating worse predictive accuracy than the historical mean.

To formally test the models' relative predictive performance, we further conduct DM tests for all model pairs within each asset class. Table 3 presents the results, where a positive test statistic signifies that the column model achieves higher predictive accuracy than the row model for the given asset and vice versa. Consistent with the out-of-sample performance results, LASSO and ENet perform best for S&P 500, whereas tree-based methods such as LGBM and XGB perform particularly well for gold and long-term Treasury bonds. However, across asset classes, the DM tests do not reveal statistically significant differences between the regularized linear and tree-based models.

Machine-learning models differ in how they process high-dimensional data based on their structural design. For example, LASSO and ENet improve forecasting accuracy by performing variable selection and by eliminating irrelevant variables and thus reducing overfitting risk. In contrast, other models enhance accuracy by shrinking the coefficients of less informative features toward zero rather than removing them from the model. As financial data typically exhibit a low signal-to-noise ratio—some key features drive most of the out-of-sample predictive performance—models capable of variable selection, such as LASSO and ENet, tend to outperform those that merely adjust coefficient weights.

Analysis of the key predictors

We investigate the significance and marginal associations of predictors to identify key variables for forecasting SPY, TLT, and GLD returns. To evaluate significance, we measure the reduction in R^2_{oos} when each predictor is set to zero while keeping all other variables constant. A large drop in R^2_{oos} implies that the predictor plays an important role, whereas small changes indicate limited influence. Figures 2–4 present the most significant variables for predicting each asset. For example, linear-based models such as Ridge, LASSO, and ENet find the Euro STOXX Index (eurstoxx) and the 30-year U.S. Treasury yield (USGov30yr) as important variables for predicting the S&P 500. Tree-based models commonly select the VIX and SPY as important predictors, while the MLP model highlights the one-month LIBOR (USDlibor1m) and the Global Surprise Index (surpind_global) as significant variables. As depicted in Fig. 3 for TLT, the 30-year U.S. Treasury yield remains an important predictor for LASSO and ENet, and VIX plays a key role in tree-based models. In Fig. 4, linear-based models identify the Euro STOXX Index as an important predictor for GLD, and tree-based models repeatedly select the EMBI credit spread (JPMEMBIspd). The MLP model chooses the MSCI ACWI Index (msciacwi) as the most important variable.

To investigate the marginal effect of individual predictors on target returns, we standardize all predictors to a range of -1 to 1. We then evaluate the impact on the target variables by adjusting one predictor at a time, keeping all others constant at their median levels. Through marginal association analysis, we explore how the target returns are functionally related to individual predictors. Figures 5–7 illustrate the marginal associations of selected predictors that are identified as influential in the earlier stages for each asset. For example, in Fig. 5, the Euro STOXX Index is positively associated with SPY, whereas the 30-year U.S. Treasury yield tends to have a negative association with SPY across most models. In Fig. 6, regarding TLT, the Euro STOXX Index only exhibits a

Table 3 Diebold–Mariano test

Panel (a) SPY												
	OLS	Ridge	LASSO	ENet	RF	LGBM	XGB	MLP1	MLP2	MLP3	MLP4	MLP5
OLS		0.66	**6.91**	**6.94**	**2.76**	**4.68**	**4.32**	**−9.45**	**−5.11**	0.66	0.25	**3.38**
Ridge			**7.55**	**7.62**	**2.63**	**4.63**	**4.23**	**−9.77**	**−5.19**	0.47	0.05	**3.28**
LASSO				**−2.12**	**−2.03**	−1.00	−0.72	**−13.69**	**−9.04**	**−4.49**	**−6.15**	**−3.50**
ENet					−1.95	−0.86	−0.61	**−13.57**	**−9.00**	**−4.39**	**−5.98**	**−3.32**
RF						1.94	**2.35**	**−13.58**	**−7.86**	**−2.56**	**−3.01**	−0.09
LGBM							0.14	**−14.56**	**−10.05**	**−4.34**	**−5.81**	**−2.70**
XGB								**−14.49**	**−10.17**	**−4.11**	**−5.17**	**−2.30**
MLP1									**6.75**	**13.86**	**12.58**	**13.57**
MLP2										**6.80**	**6.17**	**8.62**
MLP3											−0.74	**3.00**
MLP4												**4.79**
Panel (b) TLT												
	OLS	Ridge	LASSO	ENet	RF	LGBM	XGB	MLP1	MLP2	MLP3	MLP4	MLP5
OLS		**4.80**	**5.38**	**5.40**	**5.27**	**6.15**	**6.06**	**−13.42**	**−4.60**	**2.97**	**3.04**	**3.03**
Ridge			**4.77**	**4.78**	**4.67**	**5.52**	**5.45**	**−13.97**	**−5.23**	**2.23**	**2.37**	**2.35**
LASSO				0.03	−0.19	1.13	1.32	**−14.84**	**−8.04**	**−2.72**	**−3.41**	**−3.81**
ENet					−0.20	1.13	1.31	**−14.85**	**−8.04**	**−2.72**	**−3.42**	**−3.82**
RF						**2.00**	**1.99**	**−15.26**	**−7.86**	**−2.85**	**−3.83**	**−4.26**
LGBM							0.48	**−16.05**	**−9.13**	**−3.80**	**−5.34**	**−5.45**
XGB								**−15.84**	**−8.90**	**−3.82**	**−5.23**	**−5.39**
MLP1									**10.68**	**16.00**	**15.01**	**14.77**
MLP2										**8.43**	**7.12**	**6.63**
MLP3											0.46	0.40
MLP4												−0.09

Table 3 (continued)

Panel (c) GLD

	OLS	Ridge	LASSO	ENet	RF	LGBM	XGB	MLP1	MLP2	MLP3	MLP4	MLP5
OLS		1.72	**7.13**	**7.16**	1.03	**6.60**	**4.68**	**−17.78**	**−8.47**	**−3.82**	−0.96	**4.90**
Ridge			**6.94**	**6.97**	0.72	**6.36**	**4.33**	**−17.74**	**−8.53**	**−4.02**	−1.23	**4.71**
LASSO				−0.87	**−4.25**	0.13	**−2.76**	**−20.18**	**−11.34**	**−8.18**	**−6.38**	**−2.09**
ENet					**−4.24**	0.19	**−2.72**	**−20.18**	**−11.34**	**−8.16**	**−6.36**	**−2.06**
RF						**4.58**	**3.44**	**−19.24**	**−10.84**	**−4.52**	−1.85	**2.85**
LGBM							**−3.32**	**−19.96**	**−11.36**	**−8.00**	**−6.32**	**−2.05**
XGB								**−19.67**	**−10.86**	**−7.06**	**−4.98**	0.52
MLP1									**13.27**	**16.30**	**18.22**	**19.12**
MLP2										**6.03**	**8.34**	**10.54**
MLP3											**5.55**	**7.58**
MLP4												**5.78**

This table presents Diebold–Mariano (DM) test statistics comparing model performance. A positive value indicates that the column model achieves higher prediction accuracy for the corresponding asset than the row model, and vice versa. Bold values denote statistical significance at the 5% level. The corresponding *p*-values for all test statistics are reported in Appendix Table A4

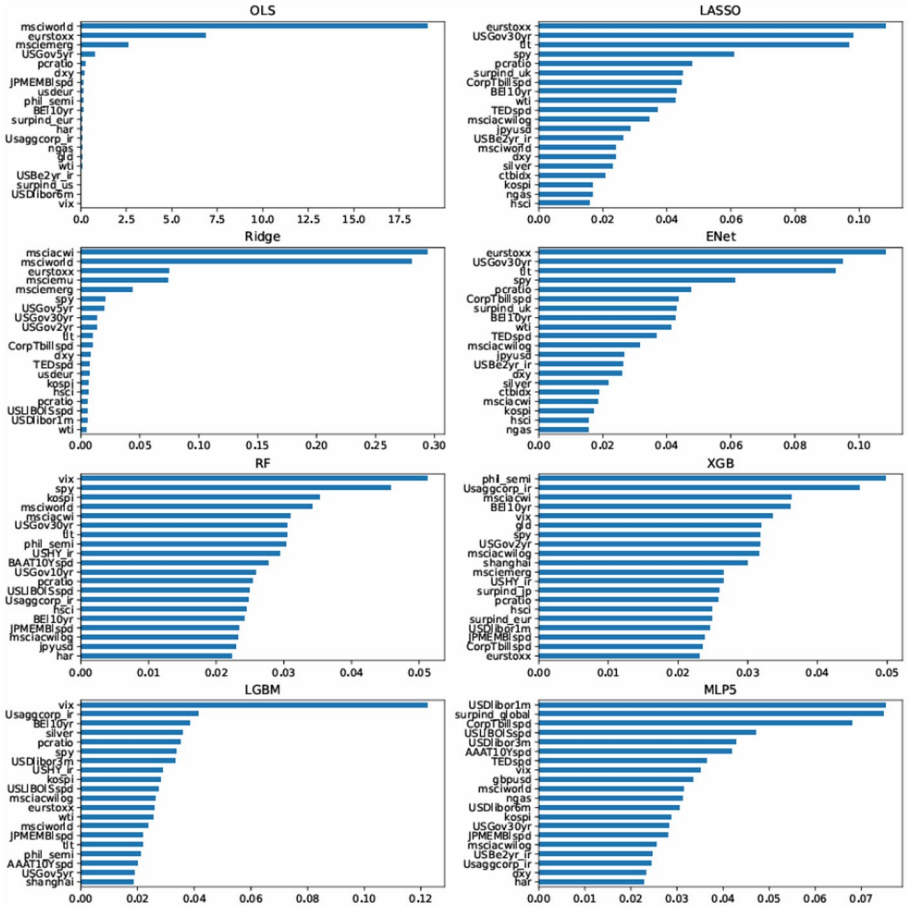


Fig. 2 SPY prediction feature importance. These graphs display the 20 most significant variables for predicting SPY returns in each model, ranked by feature importance. Feature importance is determined by the reduction in R^2 and normalized to sum to one within each model. Variables are listed in descending order of importance

positive effect in the MLP model, whereas the 30-year U.S. Treasury yield is generally associated with a positive impact. GLD is positively related to the Euro STOXX Index when the model is LASSO or ENet and to the MSCI Emerging Markets Index (msciemerg) in most models, as shown in Fig. 7.

Portfolio performance

This section explores the potential for portfolio strategies utilizing machine learning to outperform existing strategies. In mean–variance optimization, individual risky asset weights are constrained between -1 and 1.[1] The covariance matrix is calculated using daily returns from the past 20 trading days. Based on this covariance matrix and

[1] We test various alternative weight constraints, finding that optimal portfolios naturally maintain moderate positions rather than extreme allocations or excessive leverage. Our main economic findings remain robust across constraint specifications.

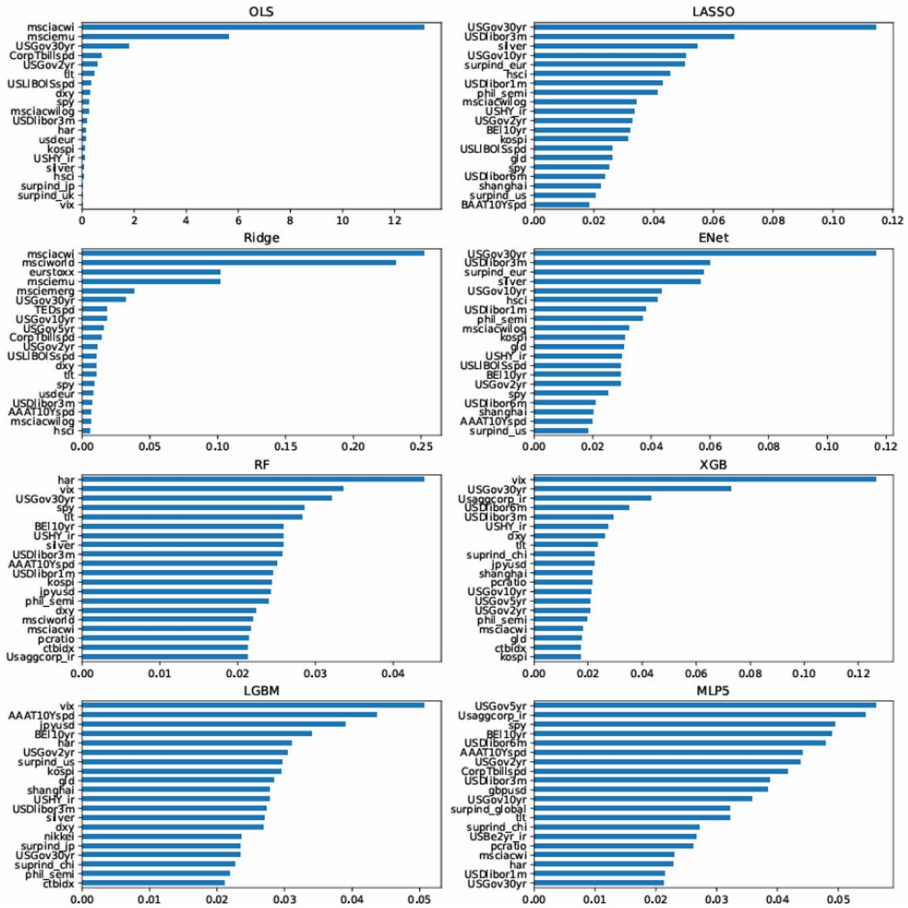


Fig. 3 TLT prediction feature importance. These graphs display the 20 most significant variables for predicting TLT returns in each model, ranked by feature importance. Feature importance is determined by the reduction in R^2 and normalized to sum to one within each model. Variables are listed in descending order of importance

predicted returns, we construct a portfolio that maximizes the Sharpe ratio. Portfolio performance is evaluated using the Sharpe ratio as follows:

$$\text{Sharpe ratio} = \frac{E\left(r_{p,t} - r_f\right)}{\hat{\sigma}_{p,t+1}} = \frac{\frac{1}{T}\sum_{t=0}^{T}\left(r_{p,t} - r_f\right)}{\sqrt{\frac{\sum_{t=0}^{T}\left(r_{p,t} - r_f\right)^2}{T-1}}}, \tag{7}$$

where $r_{p,t}$ denotes the portfolio return; r_f is the risk-free rate; and $\widehat{\sigma}_{p,t+1}$ refers to the standard deviation of the portfolio return. To provide a comprehensive risk assessment, we examine the Sortino ratio and maximum drawdown (MDD). The Sortino ratio differs from the Sharpe ratio because it considers only the downside deviation by calculating the standard deviation in the denominator of Eq. (7) using negative excess returns. MDD represents the largest peak-to-trough decline in the portfolio's value over the investment period.

Table 4 reports the performance of strategies involving the S&P 500, long-term Treasury bonds, and gold. Panel (a) presents results for machine learning-based strategies,

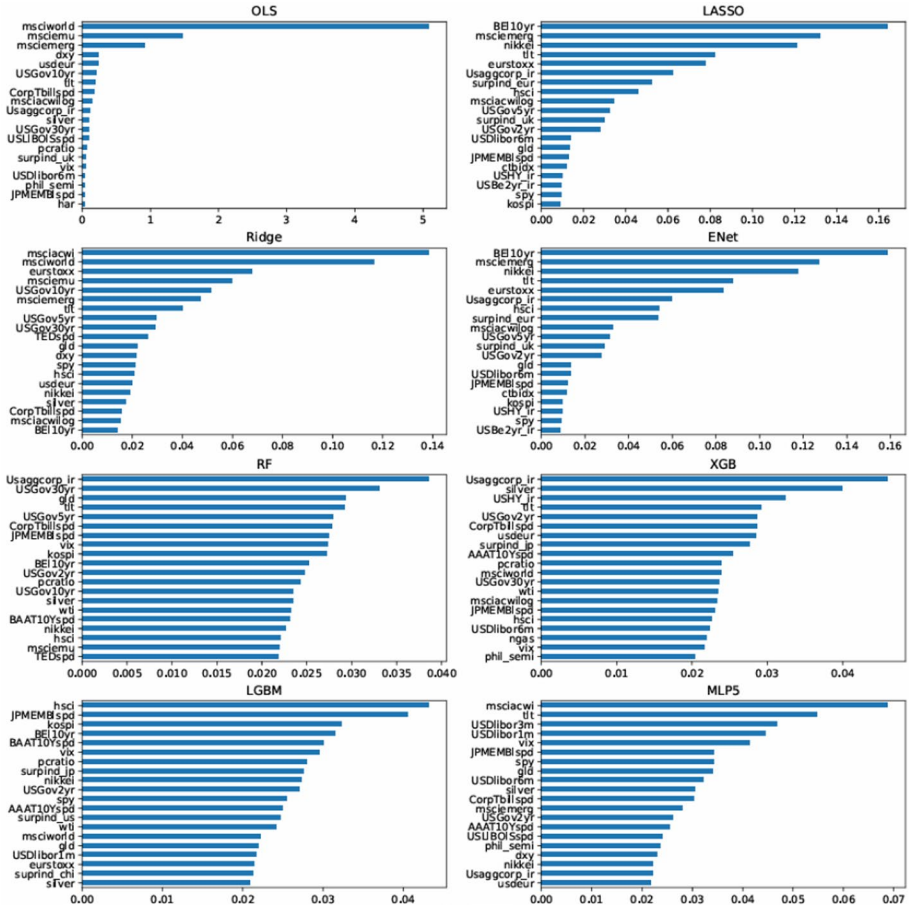


Fig. 4 GLD prediction feature importance. These graphs display the 20 most significant variables for predicting GLD returns in each model, ranked by feature importance. Feature importance is determined by the reduction in R^2 and normalized to sum to one within each model. Variables are listed in descending order of importance

while Panel (b) shows the performance of widely used traditional portfolio strategies. Traditional strategies include static asset allocation (SAA), index-tracking ETF returns, OLS, and historical mean (HM) models. The static asset allocation strategies consist of the following: (1) SAA1, which allocates equal weights to the S&P 500, long-term Treasury bonds, and gold; (2) SAA2, which assigns 30% to the S&P 500, 40% to Treasury bonds, 15% to gold, and 15% to a risk-free asset; and (3) SAA3, which distributes 30% to the S&P 500, 55% to Treasury bonds, and 15% to gold. Among the machine learning strategies, LASSO and ENet achieve the strongest performance, with annualized returns of 13.24% and 12.90%, respectively, and Sharpe ratios near 0.70.[2] These results substantially outperform the static all-weather portfolio strategies, which generate annualized

[2] The R^2_{OOS} values for SPY return prediction appear modest at 3.75% for LASSO and 3.60% for ENet (Table 2), However, Campbell and Thompson (2008) show that small predictive improvements yield significant economic gains. Following their framework and Gu et al. (2020), $SR^* = \sqrt{\frac{SR^2+R^2}{1-R^2}}$ implies a Sharpe ratio of 0.83 for LASSO compared with 0.47 for buy-and-hold SPY, demonstrating substantial economic value.

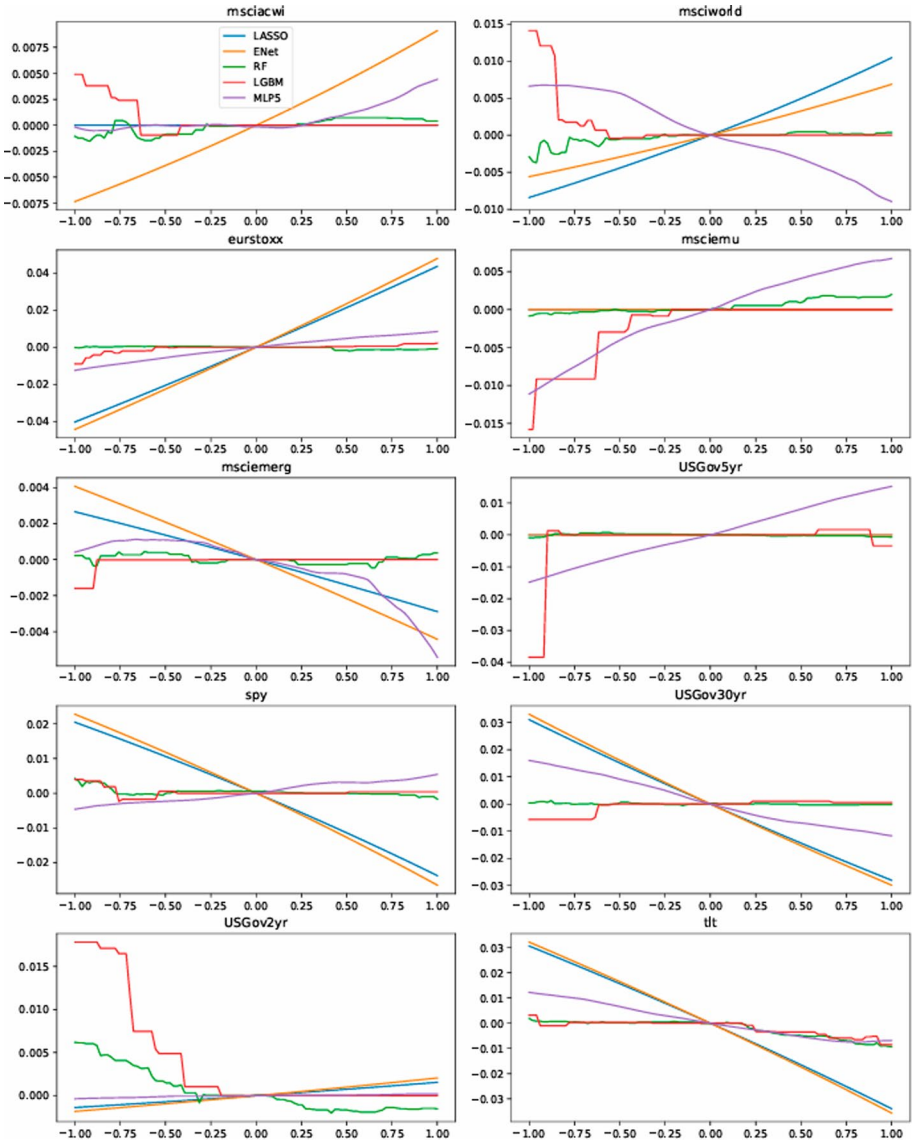


Fig. 5 SPY prediction marginal association. These figures illustrate the marginal effects of key variables on the predicted returns of SPY for each model. All predictors are standardized to the range of − 1 to 1, and the marginal effect of each is computed by changing its value while holding all other variables at their median levels

returns of 6.81% (SAA1), 5.52% (SAA2), and 5.79% (SAA3), with corresponding Sharpe ratios of 0.58, 0.55, and 0.49, respectively.

The Sortino ratio has similar values to the Sharpe ratio for most strategies, indicating that return distributions are symmetric and that performance improvements are not driven by negative skewness or excessive tail risk. Furthermore, the MDD analysis reveals that machine-learning strategies achieve substantially lower drawdowns than single-asset investments. For example, LASSO and ENet report MDDs of − 33.21% and − 34.33%, respectively, which are significantly smaller than those observed in SPY (− 55.19%) or TLT (− 48.35%). All-weather portfolios tend to exhibit smaller MDDs, aligning with their objective of delivering stable performance in response to market

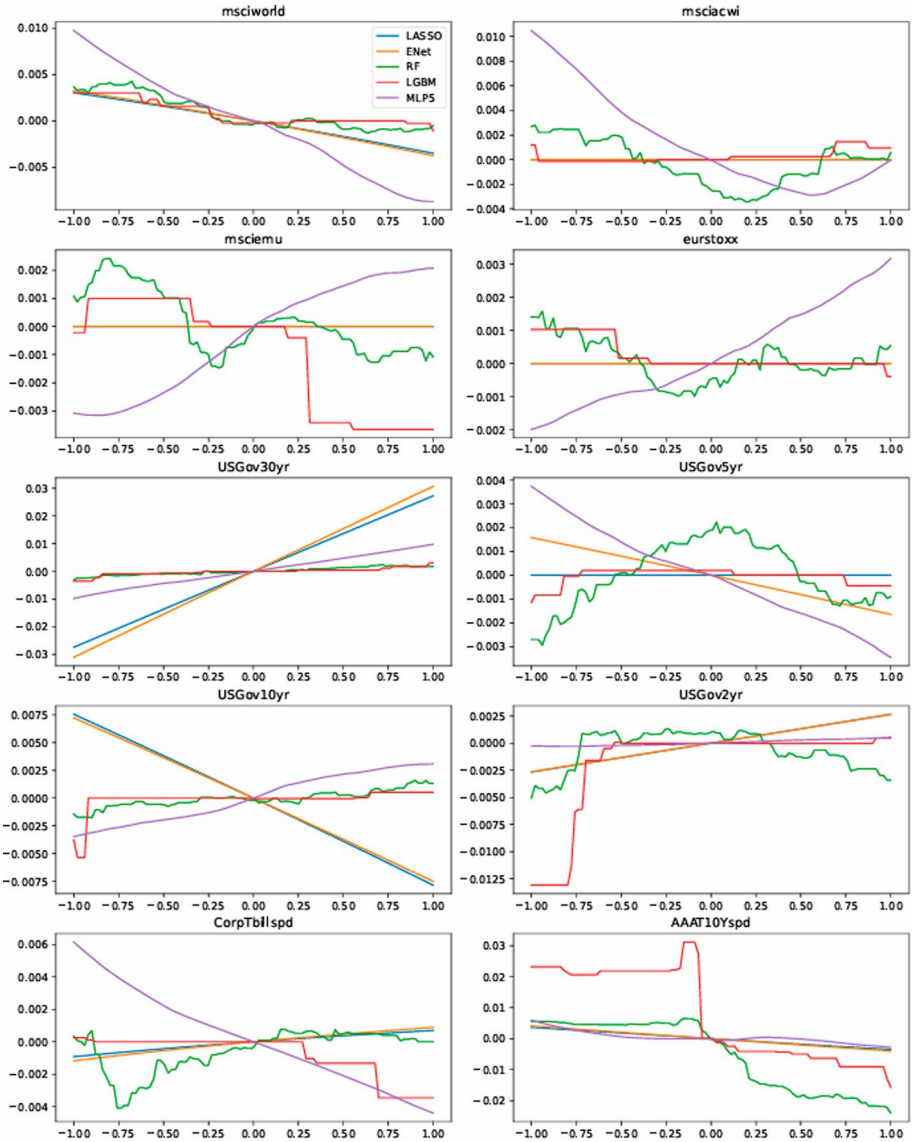


Fig. 6 TLT prediction marginal association. These figures illustrate the marginal effects of key variables on the predicted returns of TLT for each model. All predictors are standardized to the range of − 1 to 1, and the marginal effect of each is computed by changing its value while holding all other variables at their median levels

environments. As our machine learning–based strategies are designed to pursue stability and return maximization, they appear to tolerate slightly larger drawdowns as a trade-off for greater profitability, thereby achieving higher Sharpe ratios. Figure 8 illustrates the cumulative returns of representative models over our sample period, ranked by final returns. LASSO and ENet consistently exhibit superior performance, followed by random forest.

Although the results highlight the strong performance of machine learning–based strategies, their practical implementation inevitably introduces friction, most notably, frequent trading and the associated transaction costs. As documented by Avramov et al. (2023), accounting for transaction costs can significantly reduce the models' actual

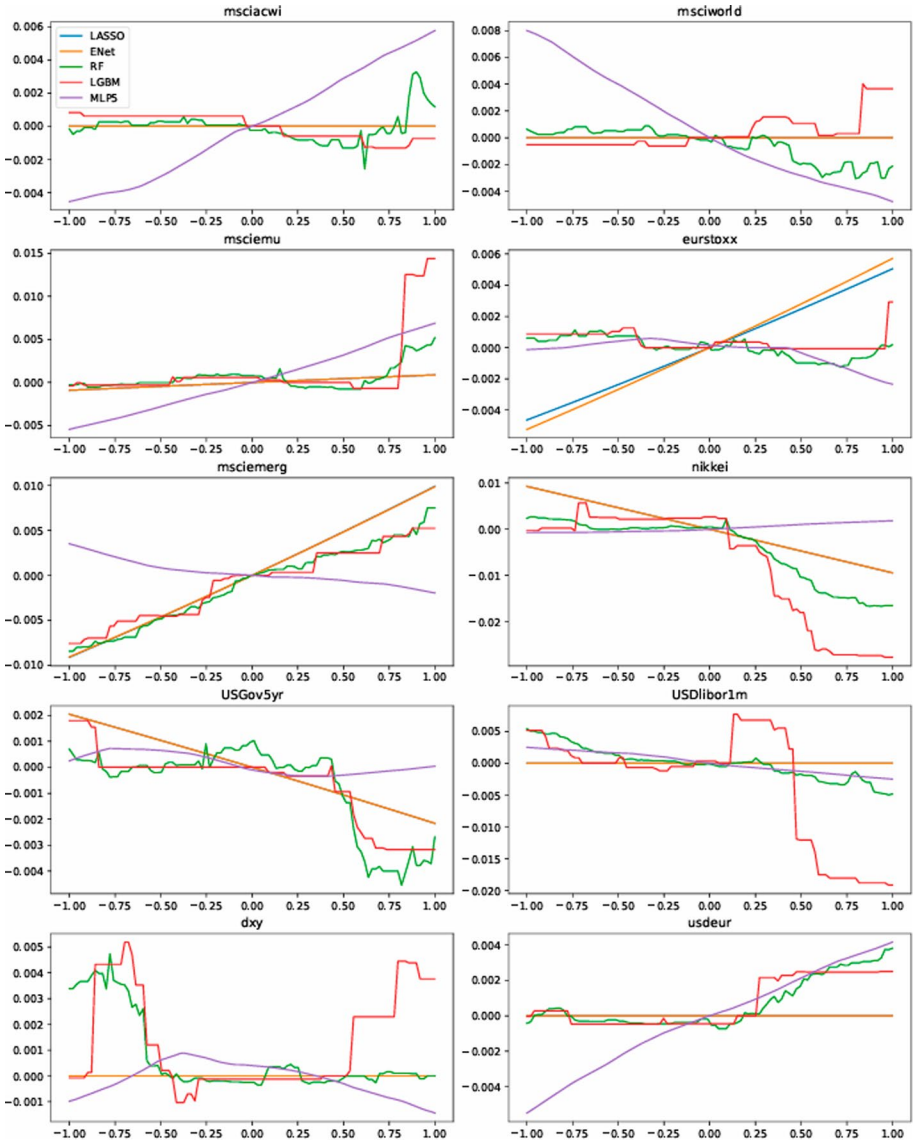


Fig. 7 GLD prediction marginal association. These figures illustrate the marginal effects of key variables on the predicted returns of GLD for each model. All predictors are standardized to the range of − 1 to 1, and the marginal effect of each is computed by changing its value while holding all other variables at their median levels

performance, with the impact particularly pronounced in extreme turnover or highly concentrated positions. To address this issue, we construct portfolios incorporating transaction costs and present the results in Appendix Table A5.[3] Even after accounting for transaction costs, LASSO and ENet continue to outperform traditional approaches. For instance, the Sharpe ratio for the LASSO strategy declines from 0.71 to 0.63 and that of the ENet strategy falls from 0.69 to 0.61; both remain higher than that of SAA1. Portfolio turnover is reported in Appendix Table A6.

[3] Compared with individual stocks, ETFs generally incur substantially lower transaction costs. For example, over the past decade, the average daily bid-ask spread for SPY has been less than 0.4 bp and under 1 bp for TLT and GLD. Regarding annual expense ratios, SPY, TLT, and GLD charge 0.09%, 0.15%, and 0.40%, respectively—all of which are minimal on a daily basis.

Table 4 Portfolio performance

Model	Return (%)	Sharpe Ratio	Sortino Ratio	MDD (%)
Panel (a) Machine Learning				
Ridge	7.75	0.34	0.31	−37.18
LASSO	13.24	0.71	0.69	−33.21
ENet	12.90	0.69	0.68	−34.33
RF	13.18	0.63	0.60	−33.96
LGBM	9.78	0.49	0.45	−48.53
XGB	13.18	0.67	0.65	−32.32
MLP1	8.59	0.43	0.39	−38.06
MLP2	11.48	0.60	0.56	−32.48
MLP3	6.66	0.30	0.27	−44.51
MLP4	5.19	0.19	0.18	−47.77
MLP5	9.77	0.39	0.38	−50.38
Panel (b) Benchmark				
SAA1	6.81	0.58	0.58	−22.74
SAA2	5.52	0.55	0.53	−22.61
SAA3	5.79	0.49	0.47	−27.73
SPY	10.56	0.47	0.45	−55.19
TLT	2.84	0.11	0.08	−48.35
GLD	7.03	0.34	0.34	−45.56
OLS	11.22	0.54	0.52	−36.77
HM	4.53	0.18	0.17	−38.78

This table reports annualized return, Sharpe ratio, Sortino ratio, and maximum drawdown (MDD) for portfolio strategies using SPY, TLT, and GLD. Panel (a) presents machine learning-based strategies, while Panel (b) displays benchmark strategies including static asset allocation (SAA), individual ETFs, OLS, and historical mean (HM). The SAA strategies represent variations of the all-weather portfolio: SAA1 equally weights all three ETFs (33.3% each); SAA2 allocates 30% to SPY, 40% to TLT, 15% to GLD, and 15% to risk-free assets; SAA3 allocates 30% to SPY, 55% to TLT, and 15% to GLD. The evaluation period spans from December 1, 2007, to July 24, 2024

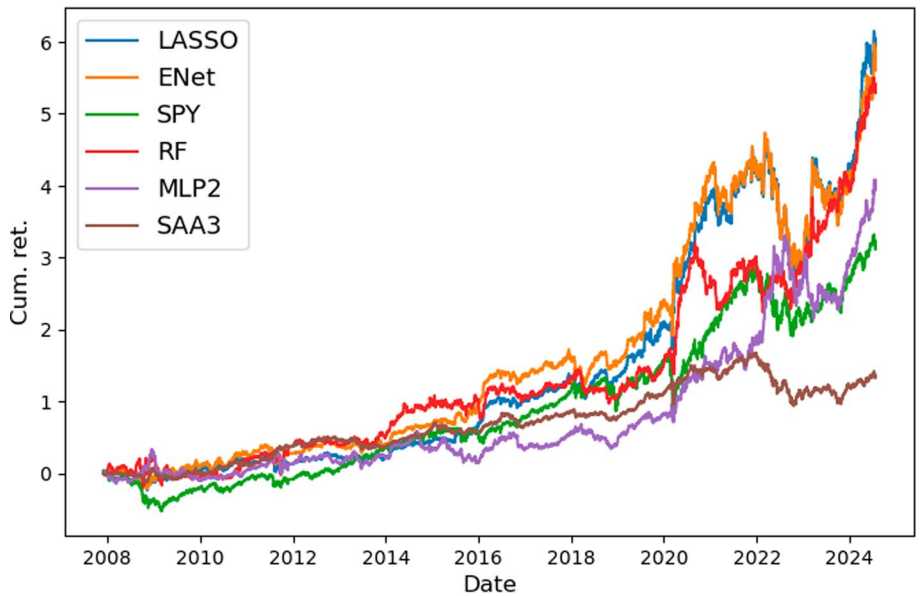


Fig. 8 Cumulative returns. This figure presents the cumulative returns of investment portfolios constructed using different models. To enable direct comparison, the legend is sorted in descending order based on final cumulative returns. The evaluation period spans from December 2007 to July 2024

Conclusion

This study examines whether machine learning can address traditional models' limitations in complex financial environments. Using a comprehensive set of financial and economic indicators, we assess the predictive accuracy of various models for the S&P 500, long-term Treasury bonds, and gold, which represent key assets in the all-weather portfolio. We find that machine learning improves predictive performance by incorporating diverse features and capturing complex interdependencies. Regularized linear models (LASSO and ENet) exhibit superior performance for S&P 500 prediction, tree-based models excel for Treasury bonds, and both approaches effectively predict gold returns. When constructing portfolios, employing LASSO and ENet achieves Sharpe ratios near 0.70, substantially outperforming static all-weather portfolios (0.58), with this advantage remaining after accounting for transaction costs.

Our findings contribute to the literature in two key ways. First, our results complement the machine learning–based portfolio strategy literature, demonstrating that machine-learning techniques can deliver superior performance to well-known SAA strategies and simple index ETFs. Second, by utilizing ETFs' liquidity, transparency, and low transaction costs, our daily frequency strategies remain competitive—even after accounting for transaction costs—demonstrating the effectiveness of machine-learning techniques in daily portfolio management. However, several limitations remain. Although we incorporate potential transaction costs, additional real-world complexities may further erode actual performance. These include market frictions such as collateral requirements, borrow fees, lot sizes, and liquidity constraints, as well as operational factors such as hidden costs, manual errors, and taxes. These issues become even more pronounced when our results are extended to stock-level portfolio management. Moreover, machine learning is not a panacea. Neural networks are prone to overfitting, and tree-based models can be highly sensitive to hyperparameter tuning, further complicating the interpretability of the machine-learning process. While our sequential validation framework maintains strict temporal separation, nested cross-validation could provide additional robustness in controlling for hyperparameter-induced optimistic bias. In this context, the set of significant predictors often varies across models, making it difficult to derive meaningful economic intuition. Addressing these complexities through advanced machine-learning and artificial-intelligence techniques with realistic constraints will be a valuable direction for future studies.

Appendix 1

See Tables Tables A1–A6

Table A1 Hyperparameter specifications

Panel (a) Linear models			
Model	Parameter	Search Grid	Search Method
Ridge	α (regularization)	5^{-5} to 5^{1}	Log-scale grid (0.1 increments)
LASSO	α (regularization)	5^{-5} to 5^{-3}	Log-scale grid (0.05 increments)
ENet	α (regularization)	5^{-5} to 5^{-3}	Log-scale grid (0.05 increments)
	l1_ratio	0.05 to 0.95	Linear grid (0.1 increments)
Panel (b) Tree-based models			
Model	Parameter	Search Grid	Fixed Parameters
Random Forest	max_depth	{7, 10, 15}	n_estimators = 300
	max_features	{10, 30, 50}	
LGBM	max_depth	{5, 10}	learning_rate = 0.01
	num_leaves	{9, 19}	n_estimators = 100
			objective = 'regression'
			boosting_type = 'gbdt'
XGB	max_depth	{3, 5, 7, 10}	learning_rate = 0.001
			min_child_weight = 1
			n_estimators = 1,000
Panel (c) Neural network models			
Model	Hidden Layers	Neurons per Layer	Fixed Parameters
MLP1	1	(512)	learning_rate_init = 0.001
MLP2	2	(512, 256)	max_iter = 10,000
MLP3	3	(512, 256, 128)	batch_size = 'auto'
MLP4	4	(512, 256, 128, 64)	early_stopping = False
MLP5	5	(512, 256, 128, 64, 32)	activation = 'relu'

This table presents the hyperparameter search space for all models

Table A2 Optimal hyperparameters from expanding window estimation

Model	**LASSO**	**ENet**		**RF**		**LGBM**		**XGB**
Estimation	**alpha**	**Alpha**	**l1_ratio**	**max_depth**	**max_ features**	**max_depth**	**num_ leaves**	**max_depth**
Panel (a) SPY								
1	0.0074	0.0074	0.85	7	30	5	9	3
2	0.0074	0.0074	0.85	15	10	5	9	3
3	0.0068	0.0074	0.85	15	10	5	9	3
4	0.0074	0.0074	0.85	7	10	10	19	3
5	0.0026	0.0058	0.45	7	50	5	9	7
6	0.0074	0.0074	0.85	10	10	5	9	5
7	0.0042	0.0058	0.75	15	10	5	9	5
8	0.0058	0.0068	0.85	7	50	5	9	7
9	0.0039	0.0046	0.85	7	10	5	9	3
10	0.0074	0.0074	0.85	7	10	5	9	3
11	0.0063	0.0074	0.85	7	30	5	9	10
12	0.0074	0.0074	0.85	7	10	5	9	10
13	0.0003	0.0004	0.65	7	10	5	9	3
14	0.0074	0.0074	0.85	7	10	10	19	3
15	0.0058	0.0068	0.85	7	10	5	9	3
16	0.0074	0.0074	0.85	7	10	10	19	3
17	0.0074	0.0074	0.85	7	10	5	9	3
18	0.0074	0.0074	0.85	7	10	5	9	3

Table A2 (continued)

Model	LASSO	ENet		RF		LGBM		XGB
Estimation	alpha	Alpha	l1_ratio	max_depth	max_ features	max_depth	num_ leaves	max_depth
Panel (b) TLT								
1	0.0074	0.0074	0.85	15	10	5	9	3
2	0.0068	0.0074	0.85	15	10	5	9	3
3	0.0014	0.0030	0.45	7	10	5	9	3
4	0.0074	0.0074	0.85	7	30	5	9	3
5	0.0053	0.0063	0.85	7	50	10	19	10
6	0.0074	0.0074	0.85	7	10	5	9	3
7	0.0004	0.0074	0.05	7	10	5	9	5
8	0.0074	0.0074	0.85	7	50	10	19	5
9	0.0074	0.0074	0.85	7	30	5	9	3
10	0.0068	0.0074	0.85	10	10	10	19	3
11	0.0074	0.0074	0.85	7	50	5	9	3
12	0.0068	0.0074	0.85	7	10	5	9	3
13	0.0026	0.0030	0.85	7	10	5	9	3
14	0.0074	0.0074	0.85	7	10	5	9	3
15	0.0074	0.0074	0.85	7	10	5	9	3
16	0.0053	0.0063	0.85	7	10	5	9	3
17	0.0063	0.0074	0.85	7	10	5	9	3
18	0.0074	0.0074	0.85	7	10	10	19	3
Panel (c) GLD								
1	0.0074	0.0074	0.85	7	10	10	19	3
2	0.0074	0.0074	0.85	7	10	5	9	3
3	0.0058	0.0068	0.85	7	10	5	9	3
4	0.0074	0.0074	0.85	7	30	5	9	3
5	0.0046	0.0068	0.65	7	30	5	9	7
6	0.0022	0.0049	0.45	7	50	5	9	3
7	0.0049	0.0058	0.85	7	30	5	9	5
8	0.0074	0.0068	0.85	7	10	5	9	3
9	0.0074	0.0074	0.85	7	10	5	9	3
10	0.0074	0.0074	0.85	7	10	10	19	3
11	0.0046	0.0063	0.75	7	50	5	9	3
12	0.0074	0.0074	0.85	7	10	5	9	3
13	0.0030	0.0068	0.45	7	30	10	19	3
14	0.0033	0.0039	0.85	15	10	10	19	7
15	0.0068	0.0074	0.85	10	10	5	9	7
16	0.0074	0.0074	0.85	7	10	5	9	3
17	0.0074	0.0074	0.85	7	50	10	19	10
18	0.0063	0.0074	0.85	7	10	10	19	3

This table presents optimal hyperparameters identified through grid search across 18 expanding window periods

Table A3 Variable descriptions

Variable	Description	Type	Frequency
Target variables			
SPY	S&P 500 ETF	Price index	Daily
TLT	Long-term treasury bond ETF	Price index	Daily
GLD	Gold ETF	Price index	Daily
U.S. economic indicators			
TEDspd	U.S. TED spread	Credit spread	Daily
LIBOROISspd	U.S. Libor-OIS spread	Credit spread	Daily
CPTBillspd	U.S. CP-TBill spread	Credit spread	Daily
BAA10Yspd	U.S. BAA-T10Y spread	Credit spread	Daily
MuniAAA10Yspd	U.S. municipal bond—T10Y spread	Credit spread	Daily
BEI10Y	U.S. 10Y breakeven inflation	Inflation expectation	Daily
BEI2Y	U.S. 2Y breakeven inflation	Inflation expectation	Daily
CorpOAS	U.S. aggregate corporate Avg OAS	Credit spread	Daily
HYOAS	U.S. High yield bond Avg OAS	Credit spread	Daily
USDlibor1m	ICE LIBOR USD 1 month	Interest rate	Daily
USDlibor3m	ICE LIBOR USD 3 month	Interest rate	Daily
USDlibor6m	ICE LIBOR USD 6 month	Interest rate	Daily
USGov2yr	U.S. Treasury 2Y yield	Interest rate	Daily
USGov5yr	U.S. Treasury 5Y yield	Interest rate	Daily
USGov10yr	U.S. Treasury 10Y yield	Interest rate	Daily
USGov30yr	U.S. Treasury 30Y yield	Interest rate	Daily
surpind_us	Citi economic surprise index – U.S.	Economic indicator	Daily
U.S. stock market variables			
VIX	VIX index	Volatility index	Daily
phlx_semi	Philadelphia semiconductor index	Equity index	Daily
cboe_pcr	CBOE put/call ratio	Market sentiment	Daily
Exchange rates			
DXY	Dollar index	Currency index	Daily
USDJPY	USD/JPY exchange rate	Exchange rate	Daily
EURUSD	EUR/USD exchange rate	Exchange rate	Daily
GBPUSD	GBP/USD exchange rate	Exchange rate	Daily
Global market indices			
eurstoxx	Euro STOXX index	Equity index	Daily
kospi	KOSPI index	Equity index	Daily
nky225	Nikkei 225 index	Equity index	Daily
msciemu	MSCI EMU index	Equity index	Daily
msciacwi	MSCI ACWI index	Equity index	Daily
msciworld	MSCI world index	Equity index	Daily
msciemerg	MSCI emerging markets index	Equity index	Daily
msciacwilog	MSCI ACWI Air Freight & Logistics index	Equity index	Daily
hsi	Hang Seng index	Equity index	Daily
Shcomp	Shanghai composite index	Equity index	Daily
Commodities			
Wti	WTI futures	Commodity price	Daily
Natgas	Natural gas futures	Commodity price	Daily
Silver	Silver futures	Commodity price	Daily
crb_index	CRB commodity index	Commodity index	Daily
Other indicators			
JPMEMBIspd	JP Morgan EMBI spread	Credit spread	Daily

Table A3 (continued)

Variable	Description	Type	Frequency
surpind_global	Citi economic surprise index—Global	Economic indicator	Daily
surpind_eu	Citi economic surprise index—Eurozone	Economic indicator	Daily
surpind_jp	Citi economic surprise index—Japan	Economic indicator	Daily
surpind_uk	Citi economic surprise index—UK	Economic indicator	Daily
surpind_cn	Citi economic surprise index—China	Economic indicator	Daily

This table presents the complete dataset employed in our empirical analysis, comprising 47 variables: 3 target assets (SPY, TLT, GLD) and 44 financial and economic indicators. The sample spans December 1, 2004, to July 24, 2024, with daily frequency observations (N = 5,126)

Table A4 Diebold–Mariano test *p*-values

Panel (a) SPY												
	OLS	Ridge	LASSO	ENet	RF	LGBM	XGB	MLP1	MLP2	MLP3	MLP4	MLP5
OLS		0.508	**0.000**	**0.000**	**0.006**	**0.000**	**0.000**	**0.000**	**0.000**	0.507	0.799	**0.001**
Ridge			**0.000**	**0.000**	**0.009**	**0.000**	**0.000**	**0.000**	**0.000**	0.636	0.957	**0.001**
LASSO				**0.034**	**0.042**	0.318	0.473	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**
ENet					0.051	0.388	0.539	**0.000**	**0.000**	**0.000**	**0.000**	**0.001**
RF						0.052	**0.019**	**0.000**	**0.000**	**0.010**	**0.003**	0.928
LGBM							0.886	**0.000**	**0.000**	**0.000**	**0.000**	**0.007**
XGB								**0.000**	**0.000**	**0.000**	**0.000**	**0.021**
MLP1									**0.000**	**0.000**	**0.000**	**0.000**
MLP2										**0.000**	**0.000**	**0.000**
MLP3											0.460	**0.003**
MLP4												**0.000**
Panel (b) TLT												
	OLS	Ridge	LASSO	ENet	RF	LGBM	XGB	MLP1	MLP2	MLP3	MLP4	MLP5
OLS		**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.003**	**0.002**	**0.002**
Ridge			**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.026**	**0.018**	**0.019**
LASSO				0.978	0.846	0.256	0.187	**0.000**	**0.000**	**0.007**	**0.001**	**0.000**
ENet					0.845	0.259	0.189	**0.000**	**0.000**	**0.007**	**0.001**	**0.000**
RF						**0.046**	**0.046**	**0.000**	**0.000**	**0.004**	**0.000**	**0.000**
LGBM							0.630	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**
XGB								**0.000**	**0.000**	**0.000**	**0.000**	**0.000**
MLP1									**0.000**	**0.000**	**0.000**	**0.000**
MLP2										**0.000**	**0.000**	**0.000**
MLP3											0.645	0.690
MLP4												0.926
Panel (c) GLD												
	OLS	Ridge	LASSO	ENet	RF	LGBM	XGB	MLP1	MLP2	MLP3	MLP4	MLP5
OLS		0.085	**0.000**	**0.000**	0.303	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	0.337	**0.000**
Ridge			**0.000**	**0.000**	0.472	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	0.218	**0.000**
LASSO				0.385	**0.000**	0.897	**0.006**	**0.000**	**0.000**	**0.000**	**0.000**	**0.037**
ENet					**0.000**	0.849	**0.006**	**0.000**	**0.000**	**0.000**	**0.000**	**0.039**
RF						**0.000**	**0.001**	**0.000**	**0.000**	**0.000**	0.064	**0.004**
LGBM							**0.001**	**0.000**	**0.000**	**0.000**	**0.000**	**0.040**
XGB								**0.000**	**0.000**	**0.000**	**0.000**	0.606
MLP1									**0.000**	**0.000**	**0.000**	**0.000**
MLP2										**0.000**	**0.000**	**0.000**
MLP3											**0.000**	**0.000**
MLP4												**0.000**

This table provides *p*-values corresponding to the Diebold–Mariano test statistics reported in Table 3. Bold values indicate statistical significance at the 5% level ($p < 0.05$)

Table A5 Robustness to weight constraints and transaction costs

Model	Weight constraint	Sharpe (gross)	Sortino (gross)	Sharpe (historical)	Sortino (historical)	Sharpe (0.5 bp)	Sortino (0.5 bp)	Information ratio
LASSO	[−1,1]	0.71	0.69	0.66	0.65	0.63	0.63	0.07
	[−0.5,1]	0.71	0.68	0.66	0.65	0.64	0.62	−0.01
	[−0.3,0.7]	0.76	0.71	0.71	0.69	0.69	0.67	−0.08
	[0,1]	0.63	0.61	0.59	0.59	0.57	0.57	−0.15
ENet	[−1,1]	0.69	0.68	0.64	0.64	0.61	0.61	0.05
	[−0.5,1]	0.73	0.71	0.68	0.67	0.65	0.64	0.01
	[−0.3,0.7]	0.81	0.78	0.76	0.76	0.74	0.73	−0.04
	[0,1]	0.61	0.59	0.57	0.57	0.54	0.54	−0.16
XGB	[−1,1]	0.67	0.65	0.61	0.61	0.58	0.59	0.07
	[−0.5,1]	0.70	0.68	0.64	0.64	0.62	0.62	0.02
	[−0.3,0.7]	0.75	0.71	0.70	0.69	0.68	0.67	−0.05
	[0,1]	0.64	0.62	0.59	0.60	0.57	0.58	−0.14
MLP2	[−1,1]	0.60	0.56	0.53	0.54	0.50	0.50	−0.01
	[−0.5,1]	0.68	0.63	0.61	0.61	0.58	0.58	−0.03
	[−0.3,0.7]	0.74	0.68	0.68	0.68	0.65	0.65	−0.08
	[0,1]	0.82	0.77	0.77	0.77	0.74	0.75	−0.02

This table reports portfolio performance under different weight constraints and transaction cost assumptions. The weight constraints are [−1,1], [−0.5,1], [−0.3,0.7], and [0,1]. Sharpe (Gross) uses no transaction costs. Sharpe (Historical) employs asset-specific half-spreads calibrated from trailing 10-year average bid-ask spreads of SPY (0.19 bp), TLT (0.43 bp), and GLD (0.39 bp). Sharpe (0.5 bp) imposes uniform 0.5 bp costs across all assets as a conservative scenario. Sortino ratios use the same cost assumptions as their corresponding Sharpe ratios. The information ratio measures risk-adjusted excess returns relative to the SPY benchmark, calculated as the mean excess return divided by tracking error

Table A6 Period-by-period portfolio turnover

Model	Weight constraint	Total	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
LASSO	[– 1,1]	254.65	188.11	331.73	280.53	144.20	214.38	260.16	156.56	101.67	130.37	92.35	84.80	236.52	370.58	444.08	497.62	439.33	407.35
	[– 0.5,1]	213.28	176.18	282.49	229.28	130.43	205.36	234.31	131.02	85.90	117.06	80.02	62.33	188.01	312.77	355.81	394.65	354.35	324.90
	[– 0.3,0.7]	154.75	145.49	199.85	165.79	99.88	160.79	188.64	104.33	65.56	79.91	44.03	45.19	133.58	225.77	256.16	285.05	239.89	210.91
	[0,1]	148.99	142.74	198.83	162.99	96.71	143.56	166.02	86.70	57.32	89.17	56.48	30.83	108.74	225.73	253.62	264.41	234.11	250.33
ENet	[– 1,1]	268.92	201.36	335.72	304.23	178.71	227.58	269.47	165.97	110.25	151.98	95.44	112.44	263.39	375.64	437.73	509.73	448.53	442.48
	[– 0.5,1]	226.97	184.74	289.79	253.56	163.21	219.67	243.77	138.11	94.65	140.81	82.68	93.69	210.41	314.79	351.53	405.55	361.59	355.65
	[– 0.3,0.7]	163.23	149.87	210.97	186.07	122.95	168.18	196.35	110.27	72.87	92.82	42.98	59.05	149.91	225.46	246.58	287.25	243.52	236.30
	[0,1]	158.49	147.60	206.33	178.20	120.64	152.90	168.67	91.13	63.38	106.79	70.61	57.10	116.41	222.05	253.41	272.74	235.54	270.56
XGB	[– 1,1]	303.36	332.62	415.82	303.49	256.50	153.31	212.38	221.64	301.46	309.80	253.93	318.97	341.36	291.21	214.14	446.90	404.97	401.53
	[– 0.5,1]	252.60	272.97	344.96	270.28	221.00	147.32	185.40	182.85	257.16	253.97	202.64	263.86	280.73	240.15	179.61	362.58	320.33	326.13
	[– 0.3,0.7]	188.01	205.62	248.48	207.36	172.24	116.85	144.95	136.02	193.53	191.86	149.69	188.54	206.51	177.39	131.08	265.53	232.46	241.30
	[0,1]	162.34	180.40	223.13	181.25	154.30	120.29	124.81	117.88	164.57	164.48	114.06	178.09	177.34	147.61	88.51	229.43	204.31	199.19
MLP2	[– 1,1]	354.15	258.43	291.89	296.19	260.54	302.57	352.41	353.97	439.81	445.16	469.51	349.79	341.60	284.02	318.59	506.83	377.64	374.20
	[– 0.5,1]	297.47	221.44	238.27	259.13	230.75	269.88	312.83	298.15	364.50	374.91	372.32	282.60	299.12	235.93	253.44	417.77	310.48	321.63
	[– 0.3,0.7]	217.13	172.13	179.98	192.93	171.15	206.20	224.48	221.38	273.75	268.26	272.82	215.60	216.60	161.76	173.82	289.79	219.59	237.21
	[0,1]	199.68	166.15	160.57	171.43	174.73	161.37	209.40	201.43	235.44	247.71	222.30	193.54	204.44	174.51	172.56	284.83	197.99	220.08

This table reports annual portfolio turnover across 17 estimation periods under different weight constraints. Annualized turnover is calculated as

$AnnualizedTurnover = 250 \times \frac{1}{T} \sum_{t=1}^{T} \left(\sum_i \left| w_{i,t+1} - \frac{w_{i,t}(1+r_{i,t+1})}{1+\sum_j w_{j,t} r_{j,t+1}} \right| \right)$,

where $w_{i,t}$ represents the weight of asset i at time t, $r_{i,t+1}$ is the return of asset i from time t to $t+1$, and the summation over j represents all assets in the portfolio. The "Total" column represents the average annual turnover across all periods. Each estimation period corresponds to one year of out-of-sample predictions with annual model retraining. Higher turnover reflects more frequent portfolio rebalancing

Acknowledgements
We would like to express our gratitude to the editor and the anonymous reviewers for their valuable suggestions that greatly improved the paper.

Author contributions
YH implemented machine learning models and data analysis. JKang designed the research methodology and portfolio analysis. JKim contributed to literature review and empirical interpretations. DC conceptualized the study and supervised the research. All authors read and approved the final manuscript.

Funding
This research was supported by the Regional Innovation System & Education (RISE) program through the Gangwon RISE Center, funded by the Ministry of Education (MOE) and the Gangwon State (G.S.), Republic of Korea. (2026-RISE-10-002).

Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests
The authors declare that they have no competing interests.

Received: 14 March 2025 Accepted: 18 March 2026
Published online: 27 April 2026

References

Aleti S, Bollerslev T, Siggaard M (2025) Intraday market return predictability culled from the factor zoo. Manage Sci. https://doi.org/10.1287/mnsc.2023.01657
Alonso Robisco A, Carbó Martínez JM (2022) Measuring the model risk-adjusted performance of machine learning algorithms in credit default prediction. Financial Innovation. https://doi.org/10.1186/s40854-022-00366-1
Avramov D, Cheng S, Metzker L (2023) Machine learning versus economic restrictions: evidence from stock return predictability. Manage Sci 69(5):2587–2619
Awan U, Shamim S, Khan Z, Zia NU, Shariq SM, Khan MN (2021) Big data analytics capability and decision-making: the role of data-driven insight on circular economy performance. Technol Forecast Soc Change 168:120766
Ban GY, El Karoui N, Lim AE (2018) Machine learning and portfolio optimization. Manage Sci 64(3):1136–1154
Ben-David I, Franzoni F, Moussawi R (2018) Do ETFs increase volatility? J Finance 73:2471–2535
Black F, Litterman R (1992) Global portfolio optimization. Financ Anal J 48(5):28–43
Bryzgalova S, Pelger M, Zhu J (2023) Forest through the trees: Building cross-sections of stock returns. Available at SSRN 3493458.
Campbell JY, Thompson SB (2008) Predicting excess stock returns out of sample: Can anything beat the historical average? Rev Financ Stud 21(4):1509–1531
Chen L, Pelger M, Zhu J (2024) Deep learning in asset pricing. Manage Sci 70(2):714–750
Chinco A, Clark-Joseph AD, Ye M (2019) Sparse signals in the cross-section of returns. J Finance 74(1):449–492
Chopra VK, Ziemba WT (1993) The effect of errors in means, variances, and covariances on optimal portfolio choice. J Portf Manag 19(2):6–11
Chun D, Kang J, Kim J (2024) Forecasting returns with machine learning and optimizing global portfolios: evidence from the Korean and US stock markets. Financial Innovation. https://doi.org/10.1186/s40854-024-00648-w
Cong LW, Tang K, Wang J, Zhang Y (2021) AlphaPortfolio: Direct construction through deep reinforcement learning and interpretable AI. Available at SSRN 3554486.
Dalio R (2017) Principles: Life and Work. Simon & Schuster, New York
DeMiguel V, Garlappi L, Uppal R (2009) Optimal versus naïve diversification: how inefficient is the 1/N portfolio strategy? Rev Financ Stud 22(5):1915–1953
Feng G, He J, Polson NG, Xu J (2024) Deep learning in characteristics-sorted factor models. J Financ Quant Anal 59(7):3001–3036
Feng G, He J, Polson NG (2018) Deep learning for predicting asset returns. Working paper, University of Chicago, Chicago.
Foroutan P, Lahmiri S (2024) Deep learning systems for forecasting the prices of crude oil and precious metals. Financial Innovation. https://doi.org/10.1186/s40854-024-00637-z
Frankfurter GM, Phillips HE, Seagle JP (1971) Portfolio selection: the effects of uncertain means, variances, and covariances. J Financ Quant Anal 6(5):1251–1262
Freyberger J, Neuhierl A, Weber M (2020) Dissecting characteristics nonparametrically. Rev Financ Stud 33(5):2326–2377
Glosten LR, Nallareddy S, Zou Y (2021) ETF activity and informational efficiency of underlying securities. Manag Sci 67:22–47
Gu S, Kelly B, Xiu D (2020) Empirical asset pricing via machine learning. Rev Financ Stud 33(5):2223–2273
Gu S, Kelly B, Xiu D (2021) Autoencoder asset pricing models. J Econom 222(1):429–450
Hoerl AE, Kennard RW (1970) Ridge regression: biased estimation for nonorthogonal problems. Technometrics 12(1):55–67
Htun HH, Biehl M, Petkov N (2023) Survey of feature selection and extraction techniques for stock market prediction. Financial Innovation. https://doi.org/10.1186/s40854-022-00441-7

Jagannathan R, Ma T (2003) Risk reduction in large portfolios: why imposing the wrong constraints helps. J Finance 58(4):1651–1683
Jiang Z, Liang J (2017) Cryptocurrency portfolio management with deep reinforcement learning. In: Paper presented at the intelligent systems conference, London, England.
Jorion P (1986) Bayes-Stein estimation for portfolio analysis. J Financ Quant Anal 21(3):279–292
Kelly BT, Pruitt S, Su Y (2019) Characteristics are covariances: a unified model of risk and return. J Financ Econ 134(3):501–524
Kelly B, Malamud S, Pedersen LH (2023) Principal portfolios. J Finance 78(1):347–387
Khandani AE, Kim AJ, Lo AW (2010) Consumer credit-risk models via machine-learning algorithms. J Bank Finance 34(11):2767–2787
Khomyn M, Putniņš T, Zoican M (2024) The value of ETF liquidity. Rev Financ Stud 37:3092–3148
Kozak S, Nagel S, Santosh S (2020) Shrinking the cross-section. J Financ Econ 135(2):271–292
Leippold M, Wang Q, Zhou W (2022) Machine learning in the Chinese stock market. J Financ Econ 145(2):64–82
Lettau M, Pelger M (2020) Factors that fit the time series and cross-section of stock returns. Rev Financ Stud 33(5):2274–2325
Lim B, Arik SÖ, Loeff N, Pfister T (2021) Temporal fusion transformers for interpretable multi-horizon time series forecasting. Int J Forecast 37(4):1748–1764
Markowitz H (1952) Portfolio selection. J Finance 7(1):77–91
Martin IW, Nagel S (2022) Market efficiency in the age of big data. J Financ Econ 145(1):154–177
Michaud RO (1989) The Markowitz optimization enigma: Is 'optimized' optimal? Financ Anal J 45(1):31–42
Rapach DE, Strauss JK, Zhou G (2013) International stock return predictability: What is the role of the United States? J Finance 68(4):1633–1662
Tibshirani R (1996) Regression shrinkage and selection via the lasso. J R Stat Soc B 58(1):267–288
Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser L, Polosukhin I (2017) Attention is all you need. Advances in Neural Information Processing Systems 30.
Wang C (2024) Stock return prediction with multiple measures using neural network models. Financial Innovation. https://doi.org/10.1186/s40854-023-00608-w
Wang C, Chen Y, Zhang S, Zhang Q (2022) Stock market index prediction using deep Transformer model. Expert Syst Appl 208:118128
Wolff D, Echterling F (2024) Stock picking with machine learning. J Forecast 43(1):81–102
Yan J, Yu J (2023) Cross-stock momentum and factor momentum. J Financ Econ 150(2):103716
Yeo I-K, Johnson RA (2000) A new family of power transformation to improve normality or symmetry. Biometrika 87(4):954–959
Zhang Q, Qin C, Zhang Y, Bao F, Zhang C, Liu P (2022) Transformer-based attention network for stock movement prediction. Expert Syst Appl 202:117239
Zou H, Hastie T (2005) Regularization and variable selection via the elastic net. J R Stat Soc B 67(2):301–320

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.